Equations and formulae — One-step · Two-step · Letters on both sides · Formulae

No calculator. Show the balancing step on each line, and check every answer. ★ = challenge question.

A · ONE-STEP AND TWO-STEP EQUATIONS

1 Solve these equations.

a $x - 7 = 12$ **b** $4x = 52$ **c** $\frac{x}{6} = 3$ **d** $x + 9 = 4$

2 Solve these equations.

a $3x - 4 = 20$ **b** $\frac{x}{5} + 2 = 6$ **c** $25 - 4x = 5$ **d** $4(x + 3) = 36$

e $\frac{2x}{3} = -8$

3 Solve $\frac{2x + 7}{3} = 5$

B · EQUATIONS FROM ANGLES AND SHAPES

4 Three angles on a straight line are $5x$, $3x$ and x. Work out the size of each angle.

5 The angles of a triangle are x, $2x + 15$ and $3x - 9$ (in degrees). Work out the size of each angle.

6 Two rectangles have the same area. One is 5 cm by $(x + 2)$ cm; the other is 3 cm by $(x + 6)$ cm. Work out the value of x.

C · LETTERS ON BOTH SIDES

7 Solve these equations.

- **a** $7m - 3 = 4m + 12$
- **b** $4(n - 1) = 2(n + 3)$
- **c** $3(x + 2) = 5(x - 2)$

8 Leo says: "I think of a number, multiply it by 4 and subtract 7. If instead I multiply it by 2 and add 9, I get the same answer." Write an equation and find Leo's number.

D · WORKING WITH FORMULAE

9 The formula $v = u + at$ gives the speed v of an object after t seconds. Work out a when $u = 4$, $t = 5$ and $v = 29$.

10

- **a** Write a formula for converting a time in minutes (m) into hours (h).
- **b** Use your formula to convert 390 minutes into hours.

11 The perimeter of a rectangle is $P = 2(l + w)$.

- **a** Work out P when $l = 7.5$ cm and $w = 4$ cm.
- **b** A rectangle has perimeter 30 cm and width 6 cm. Work out its length.

12 $3x + 2y = 24$. If x and y are both positive integers, give all the possible pairs of values of x and y.

13 A square of side x cm is cut from one corner of a rectangle 10 cm by 8 cm. The area of the shape that is left is 71 cm². Work out x.

Check: whatever you do to one side, do to the other · subtract the *smaller* letter term first · angles on a line 180°, in a triangle 180° · substitute back.